

Course: Introduction to Artificial Intelligence

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Assignment #3

Designing a Bayesian Network for a Pakistani Real-World Problem

Objective

Design and analyze a **Bayesian Network (BN)** for a real-world problem situated in the Pakistani context. The goal is to model uncertainty, causal relationships, and probabilistic reasoning using a principled approach.

Task Description

1. Problem Selection

- Choose a **real-world problem relevant to Pakistan** (e.g., traffic congestion, load shedding, public health, agriculture, education, urban safety, etc.).
- The problem must be **non-trivial** and involve uncertainty and multiple interacting factors.

2. Variable Identification

- Identify **at least 10 key variables**.
- For each variable:
 - Provide a brief description.
 - Justify why it is important to the problem.

3. Structure Design (Causal Relationships)

- Define **cause-effect relationships** between variables.
- Draw the **directed acyclic graph (DAG)**.
- Provide a short explanation for each relationship (why and how the dependency exists).

4. Probability Specification

- Assign **conditional probability tables (CPTs)** for all variables.
- Use **realistic or well-reasoned estimates** (literature, reports, intuition, or assumptions).
- Briefly justify each probability (no need for long explanations, but reasoning must be clear).

5. Implementation

- Implement the BN using **GeNIe Modeler**.
- Download link:
<https://download.bayesfusion.com/files.html?category=Academia>

6. Inference and Use Cases

- Create **at least 3 use cases**:
 - Set evidence on some variables.
 - Observe posterior probabilities of others.
- For each case:
 - Explain why it is interesting.
 - Interpret the results.
 - If results are counterintuitive, adjust probabilities and briefly discuss.

7. Originality Requirement

- You may take inspiration from existing BN examples, but:
 - Your **problem formulation and model must be unique**.
 - It must clearly reflect the **Pakistani context**.

Deliverables

- **Report (PDF)** including:
 - Problem description
 - Variables and justifications
 - BN structure and explanations
 - Probability tables (summarized if large)
 - Use cases and analysis
- **GeNIe Files (.xdsl)**
- *(Optional)* Any supplementary material (data sources, code, references)